A regularity exponent cannot have slope tending to minus infinity at infinity

Disproof of Conjecture 00000007991

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Abstract

Every real function bounded below on a tail is incompatible with an ordinary derivative tending to $-\infty$ as its argument tends to $+\infty$. A mean value theorem argument proves this directly. An optimal Hölder regularity exponent is nonnegative, so the claimed slope limit for $\alpha(p)$ at $p = \infty$ is impossible. The argument applies to every such exponent function and requires no formula for the actual optimal p-Laplace exponent.

A general calculus obstruction

Theorem 1. *Let f be a real function bounded below on a tail of the real line. Its ordinary derivative cannot satisfy*

$$\lim_{x\to+\infty} f'(x) = -\infty.$$

In particular, no eventually nonnegative real function has this ordinary derivative limit.

Proof. Suppose such a limit exists. There are real numbers L and a such that, for every $x \geq a$, both $f(x) \geq L$ and $f'(x) \leq -1$. Here and below an ordinary derivative on this tail is meant. Set

$$b = a + f(a) - L + 1.$$

Since $f(a) \geq L$, we have $b > a$. Differentiability on the tail implies continuity on $[a, b]$ and differentiability on (a, b). The real mean value theorem gives a point $c \in (a, b)$ with

$$\frac{f(b) - f(a)}{b - a} = f'(c) \leq -1.$$

Consequently

$$f(b) \leq f(a) - (b - a) = L - 1 < L,$$

contradicting the lower bound at b. □

The proof only needs the eventual slope bound $f' \leq -1$. It concerns derivatives on actual real intervals, rather than a discrete sequence or finite-difference surrogate.

Application to the exact conjecture

The source defines $\alpha(p)$ as the optimal exponent of $C^{1,\alpha}$ regularity of solutions of

$$\operatorname{div}(|\nabla u|^{p-2}\nabla u) = 0.$$

One of its explicit assertions, in both languages, is that the slope of the graph of $\alpha(p)$ tends to $-\infty$ as $p \to \infty$.

A Hölder regularity exponent is nonnegative by its definition. For the usual range $0 < \alpha \leq 1$, a local gradient estimate has the form

$$|\nabla u(x) - \nabla u(y)| \leq C|x - y|^{\alpha}.$$

An optimal exponent, whether attained or defined as a supremum of admissible nonnegative exponents, therefore cannot be negative. Only this necessary range condition is used here. In particular, the exact value, existence of an optimal endpoint estimate, and dependence on the spatial dimension are immaterial to the argument.

Apply Theorem 1 with $f(p) = \alpha(p)$ and $L = 0$ on the tail $p > 2$. It gives

$$\neg\left(\alpha(p) \geq 0 \text{ for all } p > 2 \quad \text{and} \quad \alpha'(p) \longrightarrow -\infty \text{ as } p \to +\infty\right).$$

Thus there is no candidate exponent function meeting even these two necessary parts of the conjecture. Additional PDE conditions can only restrict that already empty class of candidates.

Scope and derivative convention

This refutes the negative-infinite-slope conjunct, hence the conjunction in the source. It does not purport to compute the actual optimal exponent, validate or refute the proposed eigenvalue formula separately, or settle the behavior as $p \to 1$. No interpretation or sign assumption concerning the unspecified eigenvalue λ_0 is needed.

The asserted graph slope is interpreted as the ordinary real derivative with respect to the stated horizontal coordinate p. It is not a slope after reparametrizing the axis, nor a limit taken only along a selected discrete sequence of values of p. The theorem is about a whole tail of real arguments.

In Lean, `deriv` is a total function and equals zero at a point where the function is not differentiable. A proposed limit to $-\infty$ makes its values eventually at most -1. This *forces* differentiability at every point of that tail; the proof explicitly derives this fact and then uses the ordinary mean value theorem. No prior differentiability hypothesis is silently added to the final Lean theorem.

Formal conclusions and reproduction

The Lean project also checks the source's displayed scalar formula when its eigenvalue symbol is a fixed real parameter ℓ. Nonnegativity at $p = 3$ forces $\ell > -2$: otherwise its first branch $2/(4 + 3\ell)$ is negative. Put $c = \ell + 2 > 0$ and $A = \max(3, \ell + 5, 4/c)$. For $p > A$, both denominators are positive and

$$(p-1)^2 - (cp-2) = p(p-\ell-4) + 3 > 0.$$

Thus the minimum in the formula is exactly $1/(p-1)$ on that entire open tail. Its actual derivative is $-1/(p-1)^2 \to 0$, also incompatible with the claimed limit. This supplementary result proves eventual differentiability for the displayed constant-parameter formula itself. The main contradiction on the preceding pages does not require that formula or any assumption about whether an intended eigenvalue could depend on p.

The final theorem `conjecture_cusp_impossible` directly excludes every graph satisfying the necessary nonnegative range and the claimed ordinary slope limit. The general results use the actual real mean value theorem on the interval constructed above. A separate version takes values directly in the nonnegative reals. No PDE or special formula is built into the general function hypotheses.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The complete project is checked by a fresh build and strict Lean checking, with warnings treated as errors. Printed theorem dependencies are audited for standard foundational axioms only. There are no proof placeholders, custom axioms, or numerical computation oracles. No auxiliary numerical program is needed; the argument is an exact calculus proof.

Compile this document with `tectonic main.tex`. The submission includes the exact bilingual source, provenance, actual build logs, artifact hashes and PDF checks. An author agent developed the formal proof and supplied a self-review; the parent reviewed the proof and the final artifacts separately. No external independent review or novelty of the elementary calculus fact is claimed.

References

[1] The Last Math Competition, Conjecture 00000007991. Exact bilingual source and provenance included in this submission.

[2] The Mathlib community, *Mathlib*, revision specified above, the calculus library for real mean value theorems. The submitted source explicitly applies `exists_deriv_eq_slope` on a real interval.